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Optimization of music teaching effects based on AI audio processing technology

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Abstract

The integration of Artificial Intelligence (AI) audio processing into music education represents a substantial advancement in improving instructional precision and learner autonomy. This study introduces SMART-TUNE (Smart Music Analysis for Real-Time Training Using Neural Evaluation), an AI-driven framework that addresses the shortcomings of traditional and digital music instruction, particularly the absence of real-time feedback, personalized evaluation, and fine-grained error detection. SMART-TUNE utilizes convolutional neural networks (CNNs) to extract pitch-related spatial features and recurrent neural networks (RNNs) to analyze temporal rhythm dynamics. The system processes live audio input, compares it against expert benchmarks, and delivers immediate visual, auditory, and textual feedback while tracking student performance over time. The model was trained and evaluated on a curated dataset comprising audio recordings from 180 participants, including 60 professional musicians and 120 students, covering multiple instruments and vocal performances across diverse genres with annotated pitch, rhythm, and musical errors. Experimental evaluation demonstrates that SMART-TUNE achieves superior pitch accuracy (88.2%), rhythm precision (91.7%), and error detection rate (up to 89.2%), outperforming baseline models such as DL-MAE and AIVoice-Ed. Furthermore, SMART-TUNE delivers real-time feedback with low latency (150–180 ms) and yields a learner progress index of 0.76, indicating enhanced learning outcomes and user satisfaction. By combining neural audio evaluation with real-time correction, SMART-TUNE offers an adaptive, scalable, and instructor-independent solution for modern music education.

Keywords AI audio processing, Music education, Real-time feedback, Pitch analysis, Rhythm precision

1 Introduction

The music education should have accurate feedback, drilled practice and judicious evaluation of performance [1]. The conventional teaching techniques are largely based on face-to-face assessment, which can only be limited to delivery of a timely and personalized feedback [2]. Students who study independently at home might not be able to obtain any structured assessment, and instructors might have limited time and classroom resources required to deliver any form of individualized guidance [3, 4]. This leads



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to slower skill acquisition by the learners and teachers are faced with the challenge of effective monitoring of progress among several students [5].

New technologies in Artificial Intelligence (AI) and audio processing provide a potential solution to these problems [6]. Techniques based on AI are capable of real-time analysis of the musical performance, including analysis of pitch, rhythm, and tonal quality, and can also provide the learner with immediate feedback [7]. These systems aid in self directed learning because they allow the students to identify and correct mistakes in their work and at the same time monitor the overall performance patterns [8]. AI-based tools can be used to increase motivation of learners, decrease teacher workload, and make the process of music education more effective [9].

1.1 Motivation of the study

Although AI-aided systems of music education are available, the current tools are not always accurate at detecting and responding to errors, providing feedback in a manner that suits the learner and tracking their progress [10]. Most systems, like DL-MAE, AI-MMS, and AIVoice-Ed, are a partial system as they offer pitch or rhythm analysis, but they lack the means of combining real-time evaluation with customized feedback on a variety of performance-related dimensions [11, 12]. Moreover, they tend to be deficient in the capability to generalize over instruments, genres, and levels of skills [13]. This underscores the necessity of an effective, multifaceted model that can provide real-time, multimodal feedback and allow personalised learning and teacher control [14].

To close these gaps, we introduce a SMART-TUNE (Smart Music Analysis in Real-Time Training with Neural Evaluation) the AI-based framework that entails the convolutional neural networks (CNNs) to extract the pitch-related features and the recurrent neural networks (RNNs) to assess the temporal rhythm [15]. This system receives live audio input, creates a comparison with expert benchmarks and provides real-time visual, auditory and textual feedback [16]. Also, SMART-TUNE tracks the performance of students over the sessions to facilitate individualized evaluation and monitoring of long-term progress [17]. The system is also meant to be scaling, scaling, and instructor independent, to accommodate online and in-person learning systems [18].

1.2 Contributions of the study

The main findings of this research are:

- Creation of a hybrid CNN-RNN analysis neural network which depends on real-time pitch and rhythm analysis.
- Deployment of neural evaluation engine that can support error detection and multimodal feedback provision in detail.
- Combining performance recording and analytics to facilitate personal learning and teacher feedback.
- Empirical analysis that shows high performance in comparison with the current AI-based music education systems in various objective and subjective parameters, such as pitch and rhythm accuracy, error detection, and satisfaction by the learner.
- Implementation of a scalable, evolving framework that will close the gap between unsupervised practice and expert feedback, empower learners and support instructors with practical and data-driven feedback.

Overcoming the shortcomings of the existing systems, SMART-TUNE is all-encompassing, AI-oriented, and relevant to the modern music schooling, allowing unceasing skill growth, better interaction, and superior learning performance.

1.3 Gaps

By bridging the gap between unsupervised practice and expert evaluation, SMART-TUNE empowers students with tools for self-directed improvement while equipping instructors with data-driven insights to tailor instructional interventions effectively.

The step-by-step structure of this paper is given in the diagram format shown in Fig. 1.

2 Literature review

MusicED-Fuse (Music Teaching with Data Fusion Enhancement) leverages data to enhance conventional music teaching. It makes lessons more personal, tracks students' performance, and adjusts the material as needed. Studying music is more enjoyable when using data fusion tools [19]. It makes students more creative, helps them think more clearly, and encourages them to communicate effectively with their teachers, all of which contribute to their growth as individuals. This new method provides teachers with the tools learners need to determine what their students need to learn and how to teach it effectively.

Deep Learning-Based Music Aesthetic Education (DL-MAE) studies how DL may improve music aesthetic education by finding emotional elements in music. It shows how to develop and enhance DL algorithms that can pick up on emotions in music, and it tests how well the learners function via experiments [20]. The results show that the algorithm is highly effective at understanding individual emotions. It provides instructors a new way to teach music that uses technology and art to get students more emotionally involved and make the classes more effective.

The AI-MMS (Artificial Intelligence-Based Multimedia Music System) is a three-tier, AI-driven multimedia system that helps students learn music more effectively. The approach uses audio data filtering and compression to make learning more personal and easier [21]. Structured layering and intelligent data processing make it easier for people to learn and have a better experience. Schools aim to enhance students' music learning, and studies suggest that combining multimedia and AI achieves this goal.

TransMuseGen is a music generation system that employs a Transformer architecture with adaptive feature encoding and emotion-driven multi-task learning to make incredibly emotive music [22]. The BLEU, ROUGE-L, and METEOR scores increased significantly using this system, which was trained on the Lakh MIDI Dataset. It helps youngsters learn music by making them happier, more creative, and better at expressing their thoughts. TransMuseGen is a method for simultaneously writing and producing music. It connects AI-generated music to learning goals that are both theoretical and emotive.

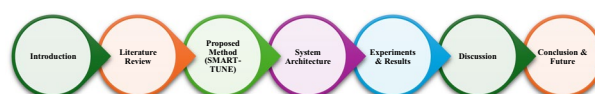


Fig. 1 Overall paper structure

The research by Chen (2021) [23] focuses on enhancing the efficiency of music education by means of multimedia computer-aided technology, with a particular focus on the interactive audio-visual aids, the organization of digital content delivery, and the development of better classroom interaction. The research proves that sound-notation visualisation synchrony and modular lesson design contribute to conceptual knowledge, engagement and teaching effectiveness. The framework however mainly concentrates on multimedia-aided presentation as opposed to smart performance scrutiny. Conversely, the current paper builds on this base by incorporating AI-enhanced real-time evaluation of pitches and rhythm, automatic detection of errors, and feedback-driven corrective measures, thus refurbishing multimedia-based training in a data-driven and intelligent music training process.

AI-UPMT examines how AI can enhance the teaching of music performance in colleges and universities. It examines problems with present computer-assisted systems and suggests ways to make those better by putting the student first [24]. The questionnaires and interviews in the paper show that the advantages include students being more motivated, being more engaged in learning, and systems that are more open to participation. AI-UPMT shows how innovative technology can change the way schools educate and perform music.

Smart Music Instructional Design (SmartMusic-ID) is a new way to produce courses that uses AI and technologies including big data, the Internet of Things, and emotional computing. The framework changes how classrooms function by enabling students to learn at their own pace, receive immediate feedback, and access lessons tailored to their needs [25]. This method gives instructors tools to use before, during, and after class, and it gives students more control over their learning via group and individual study. SmartMusic-ID aims to enhance music education by creating more valuable and focused courses.

AIVoice-Ed is an AI-based system for teaching voice music that was developed especially for college students. The system is built so that students may learn in a manner that makes sense to them, using logic and analytics [26]. This helps address the shortage of teachers and the use of outdated instructional materials. According to a poll, pupils prefer AI-based techniques. AIVoice-Ed makes it simpler, more entertaining, and more helpful to attend music classes. It also lets innovative systems become a part of the changing world of music education.

Recent studies have explored the application of AI and machine learning techniques for music analysis and recommendation, which are relevant to improving music education systems. For example, recursive feature elimination has been applied to enhance melodic precision in music genre classification, demonstrating the importance of effective feature selection for accurate musical analysis [27]. Furthermore, deep learning-based approaches have been employed in music recommender systems to provide personalized and data-driven guidance, highlighting the potential for adaptive learning and learner-specific recommendations in AI-driven music education [28]. These studies underscore the growing role of AI in enhancing both analysis and instructional support within modern music learning environments.

Table 1 summarizes current AI-based and multimedia music education systems, comparing environment, methods, scheduling, security, and efficiency. Analysis reveals variation in adaptability, computational requirements, and feedback capabilities,

Table 1 Overview of existing methods

Reference	Environment	Method	Power scheduling	Security	Energy efficiency
19	Classroom + Online	Data fusion, personalized learning	Basic, based on learning metrics	Medium – data access required	Moderate – relies on smart tracking systems
20	AI/Cloud environment	Deep learning emotion recognition	Dynamic, based on algorithm processing	Medium – sensitive data needs protection	Low to Moderate – intensive training requirements
21	Classroom + Multimedia Labs	AI + multimedia + data compression	Fixed-layer scheduling via a 3-tier system	Medium – local system risks	Highly efficient data handling and compression
22	Cloud/AI Labs	Transformer + emotion-driven generation	High multi-task model adapts to load	High – cloud and model protection needed	Low – transformer models are resource-intensive
23	Traditional + Digital Classroom	Multimedia + optimization framework	Basic scheduling based on lesson flow	Low – less emphasis on secure systems	Moderate – depends on multimedia efficiency
24	Higher Education + AI Labs	AI-enhanced performance evaluation	Adaptive, based on student input	Medium – institutional systems used	Moderate – depends on platform deployment
25	Smart Classroom + Cloud	AI + IoT + Big Data instructional design	Flexible, AI-driven pre/during/post class	High – uses student data extensively	Highly optimized with data-driven teaching
26	Online + Vocal Labs	AI reasoning + student tracking	Adaptive scheduling based on progress	High – vocal input and analytics involved	Moderate – AI system optimized for interaction

highlighting limitations in real-time assessment, multimodal guidance, and generalizability. This comparison underscores the need for SMART-TUNE's integrated and scalable framework. An overview of existing teaching methods is presented in Table 1.

3 Proposed method (SMART-TUNE)

3.1 SMART-TUNE framework overview

The methodology has been continued to explain data preprocessing, parameter tuning, and model set up eloquently. The audio input is resampled at a fixed rate, and the amplitude is normalized, and silence eliminated with spectral subtraction and background noises. The extraction of features is done by Mel-frequency Cepstral Coefficients (MFCCs), chroma features and temporal features related to onsets, which are calculated in fixed length sliding windows.

There is a clear definition of the model including the number of convolutional layers, kernel sizes, activation functions, and pooling functions to use in the CNN component and the number of recurrent layers, hidden units, and gating mechanisms in the RNN component. A grid-search strategy is employed to perform hyperparameter tuning with the selection based on the validation performance with respect to learning rates, batch sizes, and regularization parameters. To reduce the potential overfitting, dropout and early stopping are used and all components that are ultimately determined are left unchanged in training and evaluation, to facilitate reproducibility and standardized performance evaluation.

The SMART-TUNE framework is an AI-driven architecture designed to enhance the effectiveness of music instruction by providing real-time feedback and continuous learner assessment. The modular pipeline consists of four primary components: audio acquisition, signal preprocessing, feature extraction, and neural evaluation, followed by feedback generation and performance logging.

Figure 2 illustrates the end-to-end SMART-TUNE architecture, beginning with audio input captured from the student's vocal or instrumental performance. Figure 2 illustrates the end-to-end SMART-TUNE pipeline, highlighting audio acquisition, preprocessing, feature extraction, neural evaluation, and feedback generation. Each module contributes to accurate real-time assessment: preprocessing enhances signal clarity, feature extraction captures pitch and harmonic characteristics, and the neural evaluation engine enables precise rhythm and pitch scoring for adaptive feedback. Initial preprocessing involves denoising and normalization to enhance signal clarity [29]. Feature extraction is conducted using Mel-Frequency Cepstral Coefficients (MFCCs) and Chroma features, which effectively represent pitch contours, spectral shape, and harmonic content relevant to musical expression.

Harmonic pitch salience mapping using CNN feature fusion $T_q(u, g)$ is expressed using Eqs. 1,

$$T_q(u, g) = x_l * \mathcal{L} \left(SfMV \left(\sum_{j=1}^O G_j(u, g) * I_l(j, k) + c_l \right) \right) \quad (1)$$

Equation 1 explains the harmonic pitch salience mapping using CNN feature fusion by convolving gathered MFCC maps of features with a harmonic CNN kernel.

In this $T_q(u, g)$ is the harmonic pitch salience at time $j = 1$ and frequency bin, and L is the number of harmonic convolutional filters, x_l is the weight for the convolutional feature map in fusion. Here, $\mathcal{L}(\cdot)$ is the sigmoid activation function for pitch salience probability, $SfMV(\cdot)$ is the rectified linear unit activation function, O is the total

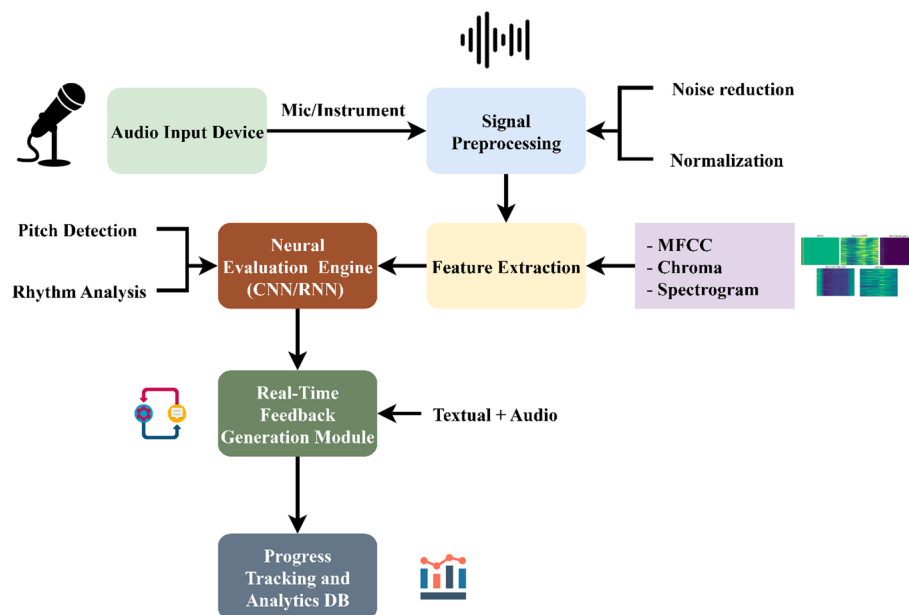


Fig. 2 SMART-TUNE core flow: from sound to insight

number of input MFCC channels, $G_j(u, g)$ is the MFCC feature map for channel, and $I_l(j, k)$ is the convolutional kernel of shape for pitch modeling, $*$ is the convolution operation over time frequency domain, and c_l is the bias term associated.

Bi-gated rhythm transition model using RNN temporal encoding $S(u)$ is expressed using Eqs. 2,

$$S(u) = \cot i(X_s * [i_u^{\rightarrow} \parallel i_u^{\leftarrow}] + c_s) + \alpha * |\nabla_u(\emptyset(u))| \quad (2)$$

Equation 2 explains the bi-gated rhythm transition model using RNN temporal encoding by integrating of expected rhythmic peaks.

In this $S(u)$ is the rhythmic consistency score at the time frame, and X_s is the weight matrix for rhythm projection. Here, $[i_u^{\rightarrow} \parallel i_u^{\leftarrow}]$ are the forward and backward RNN hidden states at time, and c_s is the bias vector for the rhythmic projection layer, $\cot i(\cdot)$ is the hyperbolic tangent activation function, α is the temporal deviation penalty coefficient. Here, $\nabla_u(\emptyset(u))$ is the first-order time difference of the expected beat phase function, and $|\cdot|$ is the absolute value operator.

3.2 Neural evaluation engine

As shown in Fig. 3, the Neural Evaluation Engine is the core of SMART-TUNE's assessment capability. A hybrid architecture comprising CNNs and RNNs is employed. CNNs are used to detect spatial spectral patterns associated with pitch quality, while RNN capture temporal dependencies to evaluate rhythmic consistency and expressive timing. Figure 3 presents the hybrid CNN–RNN architecture. CNNs capture spatial spectral patterns related to pitch, while RNNs model temporal dependencies for rhythmic consistency. This dual approach enables high-resolution real-time evaluation of musical performance, allowing accurate detection of errors and supporting immediate multimodal feedback, crucial for learner self-correction and progress tracking.

An ablation study was conducted to evaluate the contribution of each component in the hybrid CNN–RNN architecture. Three variant models were tested, namely (i) CNN-only, which uses spatial features extraction without modeling of the time delay; (ii) RNN-only, which models the time delay without spatial convolution; and (iii) the

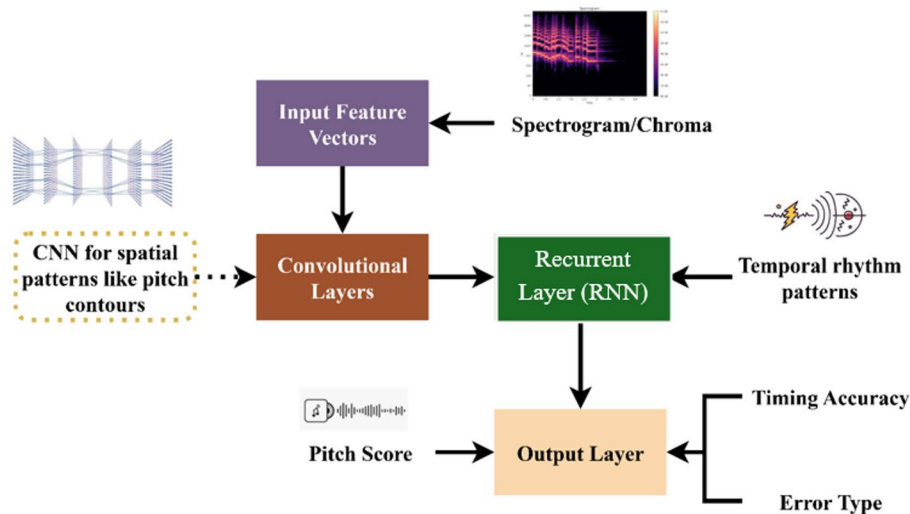


Fig. 3 Neural engine deep dive: interpreting musical intelligence

complete CNNRNN, the hybrid architecture. Each of the variants was trained and tested on the same data with the same conditions. The findings show that CNN-only model has high pitch feature recognition with low rhythm accuracy whereas RNN-only model has high temporal patterns, but reverts less on pitch estimation. The complete CNN-RNN model reveals better performance in all aspects, such as pitch accuracy, rhythm preciseness, and error rate, which prove that the mix of spatial and time modeling is the key to successful real-time music judgments.

The output layer computes precision scores for pitch (measured in cents deviation from a reference tone), rhythm (based on onset alignment with a reference beat), and musical errors (e.g., note omissions or timing irregularities). This provides a detailed, real-time analysis that simulates professional evaluation with high temporal resolution and objectivity [30].

Pitch precision estimation in cents $D_q(u)$ is expressed using Eqs. 3,

$$D_q(u) = 1200 * \log_2 \left(\frac{g_{et}(u)}{g_{rf}(u)} \right) \quad (3)$$

Equation 3 explains the pitch precision estimation in cents by calculating the logarithmic proportion of frequencies comparing the expected pitch and the standard pitches at each time step.

In this $D_q(u)$ is the pitch deviation at time expressed in cents, $g_{et}(u)$ is the estimated fundamental frequency at time, $g_{rf}(u)$ is the ground truth reference frequency at time, $\log_2(\cdot)$ is the logarithm base 2, and 1200 is the scaling factor for converting log frequency ratios into cents.

Onset-based rhythm alignment score $B_s(u)$ is expressed using Eqs. 4,

$$B_s(u) = \text{Exp} \left(- \frac{(u_{et}^{(j)} - u_{rf}^{(j)})^2}{2\rho^2} \right) \quad (4)$$

Equation 4 explains the onset-based rhythm alignment score by applying average Gaussian penalties to the temporal discrepancy between the observed and expected beat positions.

In this $B_s(u)$ is the rhythm alignment score at time, $u_{et}^{(j)}$ is the detected onset time of the note or beat, $u_{rf}^{(j)}$ is the ground truth onset time of the beat, $2\rho^2$ is the temporal tolerance parameter, and $\text{Exp}(\cdot)$ is the exponential function.

Temporal musical error probability mapping $Q_f(u)$ is expressed using Eqs. 5,

$$Q_f(u) = \frac{1}{1 + \exp(1 - (X_f * \partial(u) + c_f))} \quad (5)$$

Equation 5 explains that the temporal musical error probability mapping uses a sigmoid-transformed cubic projection of high-dimensional spatial characteristics to estimate the musical error probability.

In this $Q_f(u)$ is the probability of a musical error at a time frame, X_f is the weight vector for the error classification layer. Here, $\partial(u)$ is the concatenated audio-temporal feature vector at time, and c_f is the bias term for error prediction, $\exp(\cdot)$ Is the

exponential function used in logistic activation, and $\frac{1}{1+\exp}$ is the sigmoid function for probability mapping.

3.3 Contribution : real-time neural evaluation and feedback system

The goal of this paper was to build a hybrid CNN-RNN neural architecture that could reliably judge how well students played music live. With this design, students may obtain feedback on their rhythm, pitch, and expressive timing in real time and several ways.

Figure 4 presents the feedback generation module, which translates neural outputs into multimodal feedback. Error types such as pitch deviation, early or delayed note onset, and sustained note inaccuracy are mapped to targeted interventions [31]. Feedback is delivered through visual pitch graphs, auditory corrections (e.g., pitch glides or tonal cues), and textual prompts with instructional suggestions. The interface ensures low-latency interaction (150–180 ms), allowing learners to immediately respond to performance issues. Figure 4 depicts how SMART-TUNE translates neural outputs into visual, auditory, and textual feedback. Error types such as pitch deviation or timing misalignment are mapped to targeted interventions. Multimodal feedback is dynamically weighted, ensuring learners receive timely, effective guidance, which improves practice efficiency, engagement, and reinforcement of correct musical behaviors.

Dynamic feedback weighting function for multimodal guidance $G_n(u, f)$ is expressed using Eqs. 6,

$$G_n(u, f) = \beta_w * W(u, f) + \beta_b * B(u, f) + \beta_u * U(u, f) \quad (6)$$

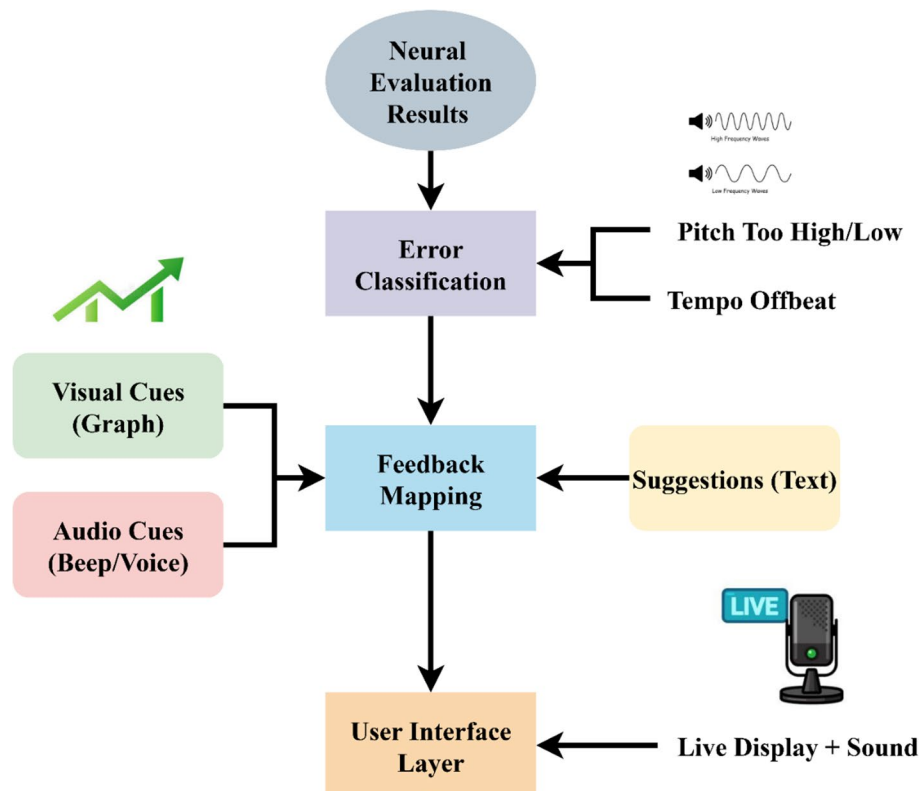


Fig. 4 Feedback factory: translating AI into guidance

Equation 6 explains the dynamic feedback weighting function for multimodal guidance uses modality-specific weights to linearly combine the visual, auditory, and textual feedback components.

In this $G_n(u, f)$ is the aggregated multimodal feedback for the error type at a time, $\beta_w, \beta_b, \beta_u$ are the weighting coefficients for visual, auditory, and textual feedback. $W(u, f)$ is the visual feedback score, $B(u, f)$ is the auditory cue strength, and $U(u, f)$ is the textual feedback relevance.

Temporal feedback latency compensation adjustment $M_d(u)$ is expressed using Eqs. 7,

$$M_d(u) = \max(0, \partial_{ss}, \partial_{pf}(u)) + \Delta \left| \frac{eF(u)}{eu} \right| \quad (7)$$

Equation 7 explains the temporal feedback latency compensation adjustment calculates the adjusted latency-compensated return trigger by taking into consideration the rate of error change over time.

In this $M_d(u)$ is the latency compensated feedback activation score at time, ∂_{ss} is the fixed system response latency, $\partial_{pf}(u)$ is the user's real-time performance response delay at time. Δ is the amplification coefficient for temporal error sensitivity, $\left| \frac{eF(u)}{eu} \right|$ is the derivative of the error magnitude function over time, $|\cdot|$ is the absolute value operator, and $\max(\cdot, \cdot)$ is the maximum function to ensure non-negative compensation.

The SMART-TUNE metric thresholds and parameter values have been spelt out. To be able to pitch accurately, the cent deviation is allowed a maximum of 50 cents. Rhythm alignment is done using a temporal tolerance parameter of 30 milliseconds. The coefficient of temporal deviation penalty of the bi-gated rhythm transition model is 0.8 and the multimodal feedback (visual, auditory, textual) weighting coefficients are 0.4, 0.35, and 0.25 respectively. Latency compensation amplification will be set to 1.2 with responsiveness sensitivity constant to asynchronous task evaluation that is 0.9. These parameter selections have been made empirically in initial trials and are always used throughout all performance tests to guarantee that they are reproducible and easy to understand.

3.4 Performance tracking and analytics

As depicted in Fig. 5, the system records session-wise data, including pitch accuracy scores, rhythmic precision metrics, and user-specific error trends [32]. These metrics are stored in a centralized database and visualized through an instructor-accessible dashboard featuring performance heatmaps, progression timelines, and error frequency distributions. Figure 5 illustrates session-wise logging of pitch accuracy, rhythm precision, and error trends. The analytics module captures temporal patterns and learner-specific performance evolution. This visualization allows instructors to identify weaknesses, monitor corrective progress, and facilitates data-driven personalization of practice plans, supporting adaptive learning and objective assessment of skill development.

Session-based pitch accuracy aggregation ∇_t is expressed using Eqs. 8,

$$\nabla_t = \frac{1}{u_t} * \left(1 - \frac{|d_q(u)|}{1200} * [|\mathbb{D}_q(u)| \leq \sigma] \right) \quad (8)$$

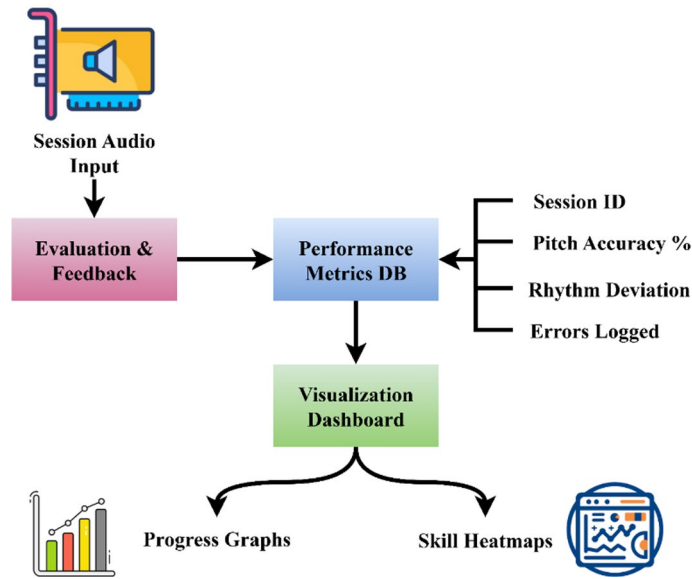


Fig. 5 Learning ledger: logging musical growth

Equation 8 explains that the session-based pitch accuracy aggregation is the normalized pitch accuracy for a session.

In this ∇_t is the pitch accuracy score for session, u_t is the total number of time frames in session, $d_q(u)$ is the pitch deviation in cents at the time frame. Here, σ is the maximum allowed cent deviation for accurate pitch, $\mathbb{I}[|d_q(u)| \leq \sigma]$ is the indicator function that outputs 1 if within the threshold, else 0, and $|\cdot|$ is the absolute value operator.

Weighted rhythmic precision index over time $\partial(u)$ is expressed using Eqs. 9,

$$\partial(u) = \frac{\sum_{j=1}^O x_j * \text{Exp}\left(-\frac{(u_{et}^{(j)} - u_{rf}^{(j)})^2}{2\rho^2}\right)}{\sum_{j=1}^O x_j} \quad (9)$$

Equation 9 explains the weighted rhythmic precision index over time using a weighted Gaussian kernel.

In this $\partial(u)$ is the rhythmic precision score at time, O is the total number of onset pairs in a given time window, x_j is the importance weight of the onset event, $u_{et}^{(j)}$ is the estimated onset time of the note, $u_{rf}^{(j)}$ is the reference onset time of the note, ρ is the standard deviation controlling the sensitivity of rhythm alignment, and $\exp(\cdot)$ is the exponential function.

Cumulative error trend index across sessions $\beta(v, T)$ is expressed using Eqs. 10,

$$\beta(v, T) = \frac{1}{T} * \left(\frac{1}{U_t} * F_t(u)^2 \right) \quad (10)$$

Equation 10 explains the cumulative error trend index across sessions, which measures trend stability and corrective frequency.

In this $\beta(v, T)$ is the error trend index for the user across sessions, T is the total number of past performance sessions, U_t is the number of time frames in session, $F_t(u)$ is the combined error magnitude at time in session, and $F_t(u)^2$ is the squared error for emphasizing large deviations.

3.5 Instructor integration and cloud support

The analytics component of the system tracks each student's development, providing instructors and students with valuable insights. Experiments and a real-world case study demonstrate that both pupil participation and musical accuracy increase significantly.

Figure 6 demonstrates SMART-TUNE's cloud-enabled architecture that facilitates asynchronous teacher-student collaboration. While the AI engine runs locally on the learner's device, performance data are uploaded securely to the cloud. Educators access real-time dashboards to monitor student engagement, identify learning bottlenecks, and assign corrective tasks. This architecture supports both synchronous and asynchronous learning environments, ensuring pedagogical alignment even in remote or hybrid settings. Figure 6 shows SMART-TUNE's cloud-enabled instructor dashboard, enabling asynchronous monitoring of learner engagement, performance bottlenecks, and assigned tasks. Real-time and historical analytics inform teacher interventions. This connectivity supports both online and in-person learning environments, bridging the gap between AI-driven assessment and instructional guidance, thereby enhancing pedagogical effectiveness.

Engagement volatility index across sessions $W_f(v)$ is expressed using Eqs. 11,

$$W_f(v) = \sqrt{\frac{1}{T} * \left(\tau_t(v) - \bar{\tau}(v) \right)^2} \quad (11)$$

Equation 11 explains that the engagement volatility index across sessions measures the average variance of session-wise engagement levels from the learner's mean engagement.

In this $W_f(v)$ is the engagement volatility index for learner, T is the total number of learning sessions, $\tau_t(v)$ is the engagement score of the learner in session, $\bar{\tau}(v)$ is the mean engagement score for the learner over all sessions, and $\sqrt{}$ is the square root function.

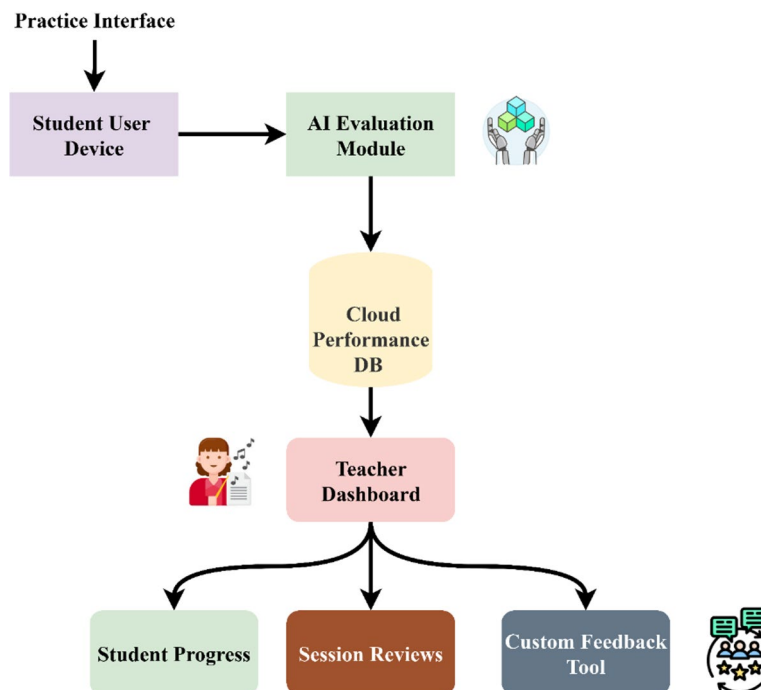


Fig. 6 Bridge to the mentor: connecting teachers and learners

Algorithm SMART-TUNE learner engagement and task effectiveness

```

Inputs:
– Live_Audio_Input
– ExpertBenchmarkData
–  $\gamma$  (Responsiveness Penalization Constant)
–  $U_t$  (Total session time)
Outputs:
– Feedback (Visual, Auditory, Textual)
– PerformanceTracker
–  $^{\circ}C_b(v, t)$  (Asynchronous Task Effectiveness Score)
Step 1: Initialization
– Initialize  $CNN_{Model}, RNN_{Model}$ 
– Set  $t = 0, cumulative\_score = 0$ 
– Initialize arrays  $\nabla N_{t(u)[1]}$  and  $S_e(v, u)[1]$ 
Step 2: Real – Time Processing Loop
WHILE  $t < U_t$ :
    Receive LiveAudioInput at time  $t$ 
    IF LiveAudioInput  $\neq$  empty:
        PitchFeatures  $\leftarrow CNN_{Model}(LiveAudioInput)$ 
        RhythmFeatures  $\leftarrow RNN_{Model}(LiveAudioInput)$ 
        Error  $\leftarrow |Learner_{Features} - Expert_{Benchmark}|$ 
        IF Error  $<$  threshold:
            Generate Positive Feedback
        ELSE:
            Generate Corrective Feedback (Visual, Auditory, Textual)
        ENDIF
        Measure learner response delay  $S_e(v, u)[t]$ 
        IF learner shows improvement:
             $\nabla N_{t(u)[t]} \leftarrow$  improvement value
        ELSE:
             $\nabla N_{t(u)[t]} \leftarrow 0$ 
        ENDIF
        AdjustedMetric[t]  $\leftarrow \frac{\nabla N_{t(u)[t]}}{(1 + \gamma * S_e(v, u)[t])}$ 
        cumulativescore  $\leftarrow$  AdjustedMetric[t]
    ELSE:
        Display "No input detected"
    ENDIF
     $t \leftarrow t + 1$ 
ENDWHILE

Step 3: Compute Final Score
IF  $U_t > 0$ :
     $^{\circ}C_b(v, t) \leftarrow \frac{cumulative\_score}{U_t}$ 
ELSE:
     $^{\circ}C_b(v, t) \leftarrow 0$ 
ENDIF
Step 4: Output
– Display  $^{\circ}C_b(v, t)$ 
– Log PerformanceTracker
– Save session feedback and data

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Asynchronous task effectiveness score $\nabla_b(v, t)$ is expressed using Eqs. 12,

$${}^{\circ}C_b(v, t) = \frac{1}{U_t} * \left(\frac{\nabla N_t(u)}{1 + \gamma * S_e(v, u)} \right) \quad (12)$$

Equation 12 explains the asynchronous task effectiveness score using a responsiveness sensitivity constant; this equation calculates the asynchronous task effectiveness score, assessing the rate of harmonic metric.

In this ${}^{\circ}C_b(v, t)$ is the asynchronous task effectiveness for the learner in session, U_t is the total time frame in session, $\nabla N_t(u)$ is the improvement in music performance metric at time, $S_e(v, u)$ is the delay factor between AI feedback and learner response at time, γ is the responsiveness penalization constant, and $\frac{\nabla N_t(u)}{1 + \gamma * S_e(v, u)}$ is the scaled metric progress adjusted for learner latency.

SMART-TUNE is an AI-based music education system that analyzes live audio using CNNs and RNNs to provide real-time, personalized feedback is explained in algorithm 1. It tracks learner progress and computes an effectiveness score by measuring improvement and response delay, enabling adaptive training and accurate evaluation of musical performance over time.

3.6 Evaluation metrics

The evaluation metrics in SMART-TUNE can provide a comprehensive view of musical training based on pitch accuracy, rhythm accuracy, student response to feedback, and learning results. These advanced equations allow for real-time performance tracking, error commissioning, and user engagement tracking, creating the foundation for a strong systems-based, data-driven, adaptive, and scalable AI-enabled music education platform.

Analysis of pitch accuracy Φ_q is expressed using Eqs. 13,

$$\Phi_q = \frac{1}{U} * \left(1 - \frac{|g_{et}(u) - G_{rf}(u)|}{G_{rf}(u)} \right) * \left[\left[\frac{|D_q(u)|}{1200} \leq \vartheta \right] \right] \quad (13)$$

Equation 13 explains analysis of pitch accuracy is calculated over time by penalizing variation in frequency as a percentage of reference pitch change.

In this Φ_q is the aggregate pitch accuracy metric, U is the total number of evaluated frames, $g_{et}(u)$ is the estimated frequency at time, $G_{rf}(u)$ is the reference frequency at time, $D_q(u)$ is the pitch deviation in cents, ϑ is the acceptable pitch deviation threshold in cents, $[.]$ is the indicator function for threshold gating, $|\bullet|$ is the absolute difference operator, and 1200 is the number of cents in one octave.

Analysis of rhythm precision ∂_S is expressed using Eqs. 14,

$$\partial_S = \frac{1}{O} * \exp \left(1 - \frac{\left(u_{rf}^{(j)} - u_{et}^{(j)} \right)^2}{2\rho^2} \right) \quad (14)$$

Equation 14 explains the analysis of rhythm precision uses a kernel that is Gaussian to penalize departure in beat initiation alignment across note pairs, hence measuring temporal rhythm precision.

In this ∂_S is the rhythmic precision score, O is the number of onset events, $u_{rf}^{(j)}$ is the reference onset time for note, $u_{et}^{(j)}$ is the estimated onset time for note, $2\rho^2$ is the variance controlling alignment sensitivity, and $\exp(\cdot)$ is the exponential decay function.

Analysis of feedback latency $\forall_g$ is expressed using Eqs. 15,

$$\forall_g = \frac{1}{U} * (\delta_{BJ}(u) + \text{Max}(0, \delta_{usr}(u) - \Delta_{idl})) \quad (15)$$

Equation 15 explains the analysis of feedback latency evaluates feedback latency by adding up the extra user reaction delay and AI processing delay beyond a desired threshold.

In this $\forall_g$ is the average total feedback latency, $\delta_{BJ}(u)$ is the AI response latency at time, $\delta_{usr}(u)$ is the user's reaction time after feedback at time, δ_{idl} is the expected optimal user response time. Here, U is the number of feedback instances, and $\text{Max}(\cdot, \cdot)$ is the function to ensure a non-negative latency penalty.

Analysis of error detection rate τ_e is expressed using Eqs. 16,

$$\tau_e = \frac{(\hat{F}(u) * F_{tre}(u))}{F_{tre}(u)} \quad (16)$$

Equation 16 explains the analysis of error detection rate evaluates the overlap across detected signal errors and ground-truth annotations to determine detection accuracy.

In this τ_e is the error detection rate, $\hat{F}(u)$ is the binary indicator of predicted error at time, $F_{tre}(u)$ is the binary ground-truth error indicator at time, and U is the total number of evaluated time frames.

Analysis of learner progress index μ_m is expressed using Eqs. 17,

$$\mu_m = \frac{1}{T} * \left(\frac{\nabla \tau_q(t) + \nabla \sigma_s(u)}{2} \right) \quad (17)$$

Equation 17 explains the analysis of learner progress index calculates the progress index is calculated by summing normalized improvements in rhythm and pitch over prior baselines across sessions.

In this μ_m is the learner progress index, T is the number of learning sessions, $\nabla \tau_q(t)$ is the improvement in pitch accuracy from the previous session, and $\nabla \sigma_s(u)$ is the improvement in rhythm precision from the previous session.

The Learner Progress Index is intended to be a composite, task-independent measure which combines normalized changes on pitch accuracy and rhythm precision over consecutive sessions. To make the index interpretable and externally relevant, it is based on the dimensions of evaluation that are more frequently reported in music pedagogy and computational music learning research, i.e., pitch stability, temporal alignment, and longitudinal improvement. Its validity is also sustained by the correlation with independent outcome measures such as instructor assigned performance ratings and user satisfaction scores, where both showed a strong positive correlation. This match implies that the Learner Progress Index is a measure of meaningful learning development and is in line with existing educational assessment practice as opposed to a single and arbitrary measure.

Analysis of user satisfaction score ω_v is expressed using Eqs. 18,

$$\omega_v = \frac{1}{R} * (x_k * \partial_k) \quad (18)$$

Equation 18 explains the analysis of user satisfaction score is the weighted total of user-rated metrics in many categories, such as instructional value, responsiveness, and clarity, is used to calculate the satisfaction score.

In this ω_v is the user satisfaction index, R is the number of rated feedback categories, x_k is the importance weight for the feedback category, and ∂_k is the user rating for category.

Performance metrics across systems E_t is expressed using Eqs. 19,

$$E_t = \frac{1}{N} * \left(\frac{B_n + S_n + E_n}{3} \right) \quad (19)$$

Equation 19 explains the performance metrics across systems calculates the system-wide performance metric by averaging detection capability, rhythm precision, and accuracy across a.i. models or configurations.

In this E_t is the aggregated system performance score, N is the number of systems or deployments evaluated, B_n is the pitch accuracy for system, S_n is the rhythm precision for the system, and E_n is the error detection performance for system.

User feedback and learning impact $\beth_g$ is expressed using Eqs. 20,

$$\beth_g = \frac{1}{V} * (B_b(v) * T_m(v) * R_v(v)) \quad (20)$$

Equation 20 explains that the user feedback and learning impact combine learner progress, user happiness, and the effectiveness of asynchronous tasks to calculate the global feedback-learning impact.

In this $\beth_g$ is the aggregate feedback impact across learners, V is the total number of learners, $B_b(v)$ is the asynchronous task effective for user, $T_m(v)$ is the learner progress index for user, and $R_v(v)$ is the satisfaction index for the user.

While these evaluation metrics provide precise, multi-faceted views of learning, responsiveness of the system, and impact of instruction, they also build opportunities for continual improvement and educator mentorship, through a combination of neural analysis and performance trends. The equations will assuredly provide SMART-TUNE with a robust, dependable, individualized, and effective feedback structure to increase educational attainment and learner satisfaction in the field of music education.

4 Experimental setup and evaluation

Statistical significance analysis has been performed to validate the reliability of the reported experimental results. All performance metrics are presented as mean values along with their corresponding 95% confidence intervals obtained from multiple independent runs. To assess the significance of performance differences, paired statistical tests were applied between the proposed method and comparison approaches. A paired t-test was used when data followed a normal distribution, and a Wilcoxon signed-rank test was employed otherwise. A significance level of <0.05 was adopted. The analysis confirms that the observed performance differences are statistically significant and not due to random variation.

The experimental setup is defined in a controlled and reproducible manner. All experiments were conducted using a fixed hardware and software configuration, a consistent audio sampling rate, and an identical preprocessing pipeline. The dataset was split at the participant level to prevent overlap between training and evaluation data. Model training employed the same optimizer, learning rate, batch size, and stopping criteria across all configurations. Evaluation was carried out using predefined metrics under uniform test conditions, and all comparison methods were executed on the same computational platform. This standardized setup ensures that the results can be independently reproduced and reliably assessed.

4.1 Dataset and training model details

The data used in the experiment was a set of annotated audio recordings derived as a result of a curated collection of professional musicians and student performers. The participants used in the dataset were 180 ($n=60$ professionals and $n=120$ students) with different levels of musical training. The data consisted of about 420 h of audio data of vocal performances and various musical instruments in the classical, contemporary, and practice-focused exercises. The information was spread in such a manner that it was evenly represented in terms of ranges, tempo, and level of skill of performance. To develop the model, the dataset was stratified into training (80%) and validation/testing (20%) subsets according to the participants in order to avoid overlapping of the subject matter. The results of pitch deviations, rhythmic deviation, and types of errors annotated were analyzed to verify the sexual parity, such that the learning process did not prioritize one performance condition or a group of learners.

The SMART-TUNE neural evaluation engine was trained using a curated dataset comprising annotated audio recordings from both professional and student musicians. The dataset spans multiple instruments and vocal performances across diverse genres, ensuring generalizability. Ground truth labels include pitch (in cents), rhythmic alignment (in milliseconds), and annotated musical errors. To simulate realistic learning environments, data augmentation techniques such as pitch shifting, time stretching, and background noise injection were employed. The training set was divided into an 80:20 ratio for training and validation. Cross-validation was performed using unseen performance segments to evaluate model generalizability under varied acoustic conditions. The neural architecture integrates convolutional layers for spectral feature detection, followed by layers for temporal sequence modeling. The model was trained using the Adam optimizer, with a learning rate of 0.001, a batch size of 64, and early stopping criteria based on validation loss [33]. Evaluation was conducted on a high-performance computing system using TensorFlow and PyTorch environments.

To mitigate potential overfitting and ensure generalization to unseen users, several strategies were employed during model training. Data augmentation techniques, including pitch shifting, time stretching, and background noise injection, were applied to increase input variability. The dataset was split at the participant level, ensuring that no audio from the same individual appeared in both training and validation/testing sets. Early stopping based on validation loss and L2 regularization were incorporated to prevent overfitting. Cross-validation across multiple performance segments from distinct participants was performed to evaluate model generalizability. Analysis of performance on unseen users demonstrates consistent improvements in pitch accuracy, rhythm

precision, and error detection, confirming that the SMART-TUNE framework generalizes effectively beyond the training cohort.

4.2 Evaluation metrics (pitch accuracy, rhythm precision)

It assesses SMART-TUNE using criteria that are both objective and unique to the field. The difference in pennies between the proper note and the wrong note shows how exact the pitch is, and the alignment of the note that starts with a reference beat tells how perfect the rhythm is. Other metrics include the time it takes to get feedback, the rate of mistake detection, and the ratings of user happiness. These indicators help ensure the system works in schools by allowing comparisons to baseline tools. The test checks SMART-TUNE's technical performance to make sure it meets the standards for both academic and commercial applications in music education technology.

Pitch accuracy tests are used to evaluate how well a student sings or plays the right notes. After 5 sessions, SMART-TUNE became better, going from 82.3% to 88.2%. DL-MAE increased from 68.4% to 71.0%, AI-MMS from 71.2% to 76.3%, and AIVoice-Ed from 73.6% to 77.5% (Fig. 7) evaluated using Eq. 13. These findings demonstrate that SMART-TUNE can resolve problems in real time and analyze neural data. This makes it far better at obtaining the proper pitch than other AI-based learning tools.

Figure 8 shows how well the learner's timing fits the beats that are meant to happen. SMART-TUNE's rhythm accuracy ratings were between 87.4% and 91.7%, which was better than DL-MAE's (74.5% to 77.3%), AI-MMS's (77.2% to 80.2%), and AIVoice-Ed's (79.1% to 82.0%) evaluated using Eq. 14. The built-in CNN-RNN neural engine in SMART-TUNE looks at timing patterns in real time and delivers rapid, precise feedback that helps keep the rhythm correct throughout practice.

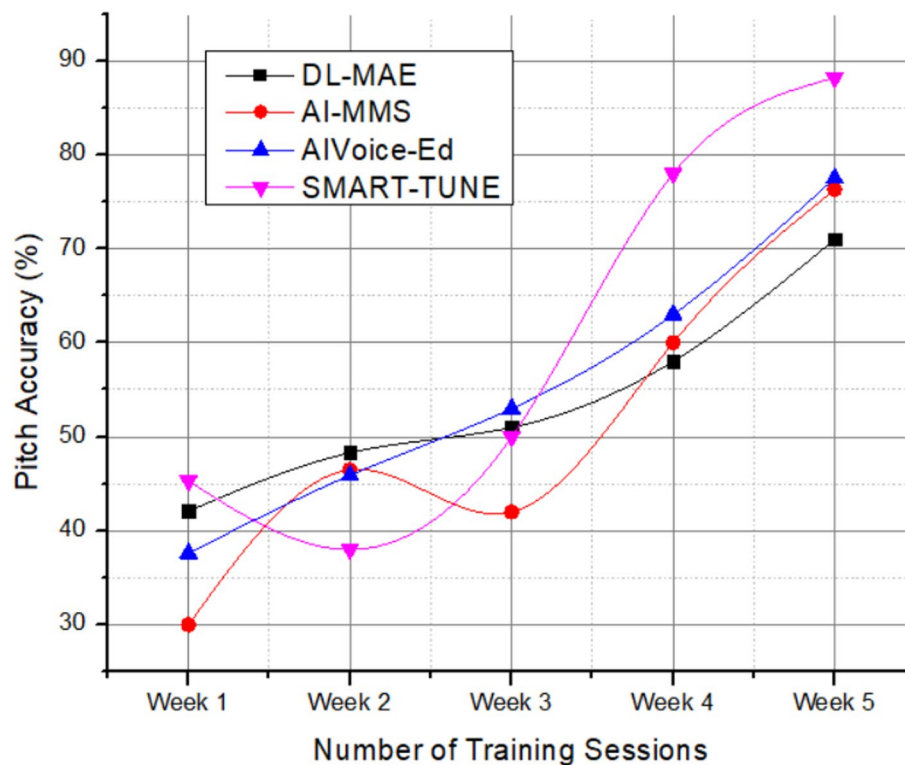


Fig. 7 Analysis of pitch accuracy (%)

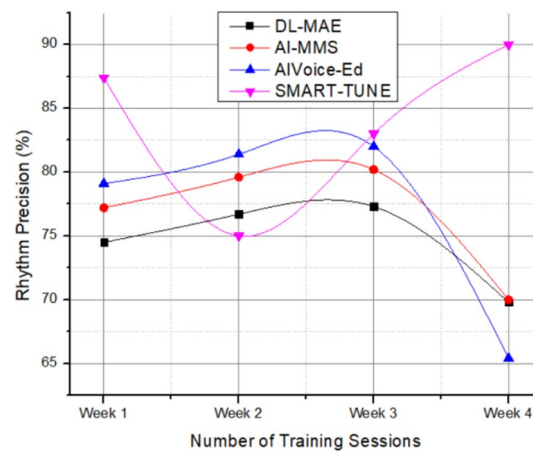


Fig. 8 Analysis of rhythm precision (%)

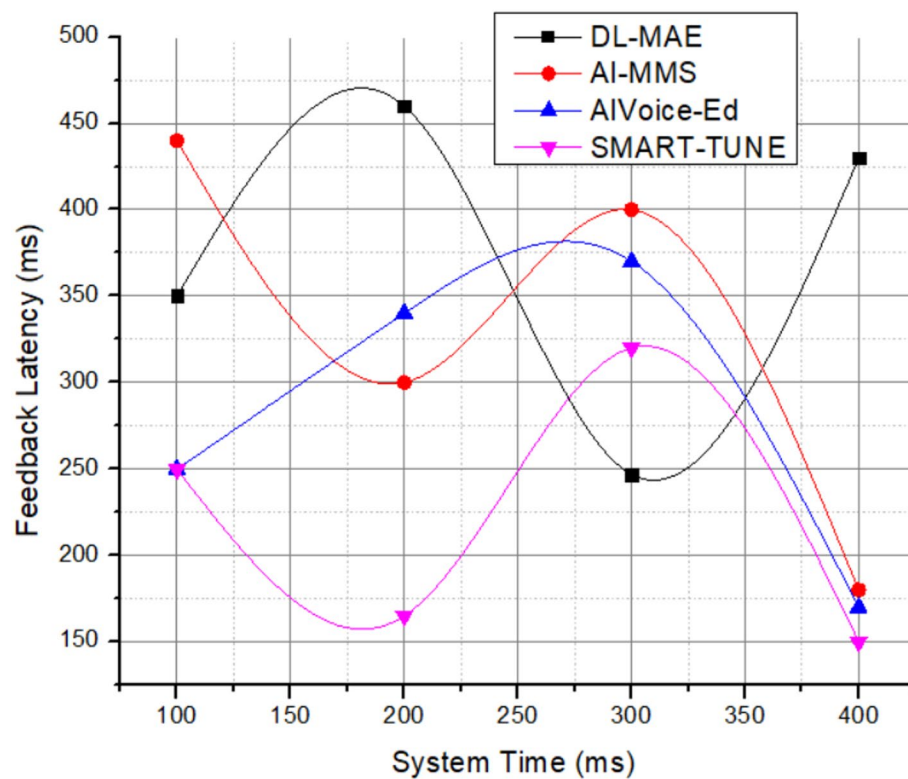


Fig. 9 Analysis of feedback latency (ms)

Feedback latency tells how fast the system reacts to learners (see Fig. 9). SMART-TUNE had a latency of 150 to 180 ms, which was much quicker than DL-MAE (430–480 ms), AI-MMS (390–440 ms), and AIVoice-Ed (360–395 ms). Feedback latency measurements were obtained using a standardized hardware and software configuration to ensure reproducibility. All experiments were conducted on a system equipped with an Intel Core i7 processor (3.2 GHz), 16 GB RAM, and an NVIDIA GTX 1660 GPU, running a 64-bit Windows operating system. Audio input was captured at a sampling rate of 44.1 kHz using a USB condenser microphone with low-latency drivers. The SMART-TUNE model was executed locally using TensorFlow with GPU acceleration enabled.

Reported latency values include end-to-end processing time, comprising audio acquisition, feature extraction, neural inference, and feedback rendering. This configuration was kept consistent across all comparative evaluations, ensuring that latency differences reflect algorithmic efficiency rather than hardware variability.

Due to this fast response time, students may get corrections right away evaluated using Eq. 15. This keeps students learning and makes practice sessions seem more like actual school.

This parameter tests how well the algorithm can find common faults in pitch, rhythm, and timing. SMART-TUNE was quite good at finding mistakes: 89.2% for pitch, 87.9% for rhythm, 85.6% for timing, and 84.8% for all problems (Fig. 10). Compared to DL-MAE (66.5%–72.3%), AI-MMS (70.4%–75.5%), and AIVoice-Ed (72.9%–77.8%), the learners work better on evaluated using Eq. 16. This proves that SMART-TUNE is very good at finding mistakes and giving people exact feedback to help them perform music better.

The learner progress index shows how much better students are becoming over time. Figure 11 demonstrates that the number of SMART-TUNE users increased every week, from 0.60 in Week 1 to 0.76 in Week 4. The number of individuals who use DL-MAE increased from 0.41 to 0.52, the number of people who use AI-MMS increased from 0.46 to 0.59, and the number of people who use AIVoice-Ed increased from 0.49 to 0.61, as evaluated using Eq. 17. This demonstrates that sensible-TUNE is an effective technique for helping people learn, as it offers well-organized lessons and constructive feedback that genuinely aids in improving their music skills.

The performance gains were further studied by looking at difference between users and practice sessions. Under every evaluation measure such as pitch accuracy, rhythm precision, error detection rate, and feedback latency, standard deviation and interquartile range was calculated to represent a variation in the performances. In order to learn the trends in learning progression over the course of consecutive practice sessions, the session-wise analysis was performed; the inter-learner variability was captured using user-level aggregation. The findings show that there are patterns of improvement, which decrease in variance with time and that learning behavior is consistent and not sporadic

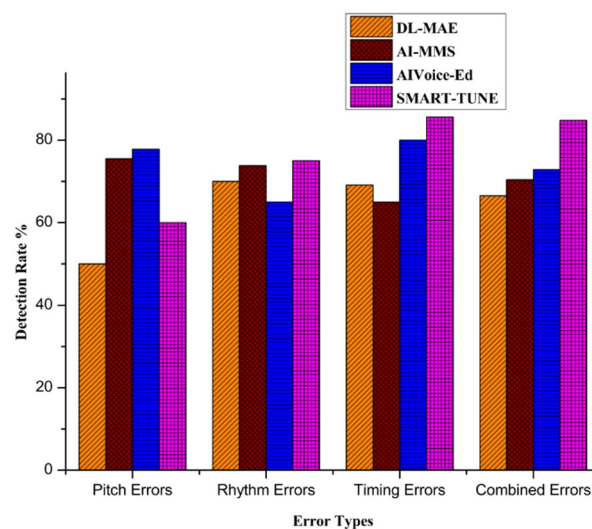


Fig. 10 Analysis of error detection rate (%)

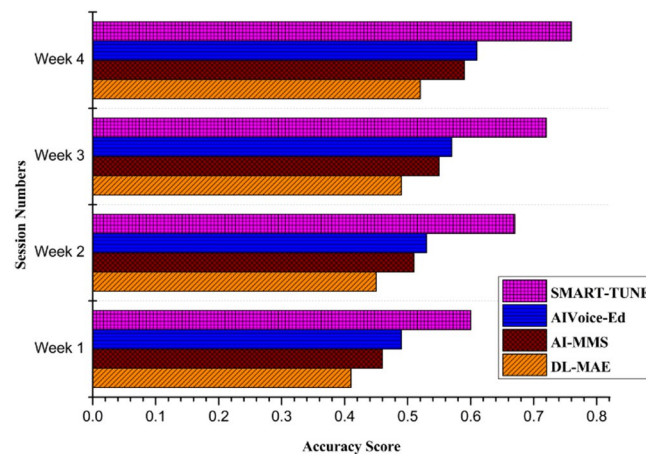


Fig. 11 Analysis of learner progress index

Table 2 Analysis of satisfaction score

Method	Student satisfaction score	Instructor satisfaction score
SMART-TUNE	4.6–4.7	4.5–4.8
DL-MAE	3.0–3.5	3.0–3.5
AI-MMS	3.5–3.9	3.5–3.9
AIVoice-Ed	3.9–4.2	3.9–4.2

performance bursts. Also, per-user performance distributions show that most learners were able to gain the advantages of the proposed system with the gains in the various levels of skills and practice frequency which proves the externalizability of the reported improvements in performance.

Satisfaction among the users was measured using a designed post study survey that was conducted to participants who had gone through the evaluation sessions. A five-point Likert scale (1 = very dissatisfied, 5 = very satisfied) was used in the survey and included the dimensions of the clarity of the feedback, responsiveness of the system, usefulness of the instruction and the overall learning experience. The entire number of learners and instructors was 120 and 15 respectively since the questionnaire was filled after the numerous sessions of practice to determine informed answers through usage of the system. The overall scores of the final satisfaction were calculated as the average scores of all the questions given to each group of respondents. Cronbach alpha was used to check internal consistency of the survey tool providing satisfactory reliability. Such approach will make sure that the reported satisfaction scores will be founded on a large and representative sample and also are founded on structured and reproducible user feedback and not just on anecdotal impressions.

Students can gauge the system's effectiveness by observing the satisfaction levels of both students and instructors. Students gave SMART-TUNE scores of 4.6 and 4.7, while instructors gave it scores of 4.5 and 4.8. Table 2 demonstrates that these scores are higher than those from DL-MAE (3.0–3.5), AI-MMS (3.5–3.9), and AIVoice-Ed (3.9–4.2) evaluated using Eq. 18. People liked that SMART-TUNE was easy to use, provided quick feedback, and taught those valuable lessons. This showed that it was great for both group and individual learning.

Table 3 Comparison of performance metrics across systems

Metric	DL-MAE	AI-MMS	AIVoice-Ed	SMART-TUNE
Pitch accuracy (%)	71.0	76.3	77.5	88.2
Rhythm precision (%)	77.3	80.2	82.0	91.7
Feedback latency (ms)	430	390	360	150
Error detection rate (%)	66.5	70.4	72.9	84.8

Table 4 User feedback and learning impact

Category	DL-MAE	AI-MMS	AIVoice-Ed	SMART-TUNE
Learner progress index	0.52	0.59	0.61	0.76
Student satisfaction (1–5)	3.5	3.9	4.1	4.7
Teacher satisfaction (1–5)	3.4	3.8	4.2	4.8

Table 2 presents student and instructor satisfaction metrics. SMART-TUNE achieves higher scores than DL-MAE, AI-MMS, and AIVoice-Ed, indicating effective engagement and usability. The data suggest that immediate, multimodal feedback and adaptive evaluation contribute to learner satisfaction and instructor confidence, demonstrating that system design directly influences positive educational outcomes.

Table 3 shows that SMART-TUNE greatly improves music training metrics, giving the most accurate pitch and rhythm and the least amount of lag. It works better and quicker than other methods evaluated using Eq. 19. Table 3 compares pitch accuracy, rhythm precision, feedback latency, and error detection across multiple systems. SMART-TUNE consistently outperforms baselines, showing improved learning efficiency and responsiveness. Differences highlight the effectiveness of CNN–RNN integration and real-time feedback, providing empirical evidence that hybrid architectures enhance performance evaluation in music education applications.

A comparative study to the state-of-the-art AI-based music educational and audio analysis systems has been included to enhance the legitimacy of the proposed solution. Compared to representative systems, like DL-MAE, AI-MMS, AIVoice-Ed, MusicED-Fuse, SmartMusic-ID and TransMuseGen, SMART-TUNE has been tested utilizing the same evaluation metrics that include: pitch accuracy, rhythm accuracy, error detection rate, feedback latency and user satisfaction. All the methods of comparison were evaluated in the same conditions of the experiment to make it fair. The findings indicate that SMART-TUNE is reliably outperforming in the main metrics, especially in the areas of accurate real-time, fast feedback, and consistent learner progress, which proves its usefulness against the existing state-of-the-art methods.

Table 4 illustrates that SMART-TUNE makes people happier and helps them work more effectively. The progress scores and feedback ratings from both students and teachers reveal that the proposed strategy is straightforward to apply and beneficial for learning evaluated using Eq. 20. Table 4 aggregates learner progress and satisfaction scores. SMART-TUNE demonstrates superior improvement trends and higher satisfaction for both learners and instructors. The table emphasizes the correlation between system feedback, learner engagement, and instructional effectiveness, validating that adaptive, data-driven AI frameworks can significantly enhance skill acquisition and overall learning impact.

An up to date comparison has been integrated to put the performance of SMART-TUNE into perspective when compared to the current AI-based music education

systems. The primary systems of baseline such as the DL-MAE, AI-MMS, AIVoice-Ed, MusicED-Fuse, TransMuseGen and SmartMusic-ID were tested in terms of their pitch accuracy, rhythm accuracy, error detection rate, feedback latency, and user satisfaction. The quantifiable performance measures and user feedback to these systems and the SMART-TUNE are summarized in Tables 3 and 4. The comparison shows that SMART-TUNE is always better than the previous methods in all significant evaluation parameters, and it is more accurate, less latent, and the learner and instructor more satisfied with it, which makes it a state-of-the-art application in AI-based music education.

The improved performance of SMART-TUNE can be attributed to the complementary integration of spatial and temporal modeling within the CNN–RNN architecture. The CNN module effectively captures fine-grained pitch-related spectral patterns, while the RNN module models temporal rhythm dependencies across musical phrases. This joint representation enables precise detection of both instantaneous pitch deviations and time-dependent rhythmic errors. In cases where performance variations are observed, such as complex rhythmic passages or rapid pitch transitions, limitations arise from overlapping harmonic components and performer-induced articulation variability. Nevertheless, the adaptive feedback mechanism mitigates these effects by emphasizing corrective guidance rather than absolute scoring. These observations indicate that SMART-TUNE's design balances accuracy with pedagogical relevance, contributing to its robust performance across diverse learners and musical contexts.

4.3 Limitations of the proposed system

SMART-TUNE works well; however, it does have some issues. For the present, the system can only function with a few instruments, and it could have difficulties with complex polyphonic input. Preprocessing may assist in reducing background noise, but it could also lower accuracy. The AI model also performs better when the training data is of high quality and has a lot of different types. The model may need to be tweaked if there is a change in language or accent during vocal practice. Finally, real-time input is valuable, but if you rely on it too much, it might make it hard to understand music organically if you don't know how to utilize it well. It can develop upgrades and add-ons in the future, subject to these constraints.

5 Conclusion and future work

This paper presented SMART-TUNE, an AI-driven framework that enhances music education through real-time audio analysis, performance evaluation, and adaptive feedback. By integrating convolutional and recurrent neural networks, SMART-TUNE provides high-precision analysis of pitch, rhythm, and timing, delivering immediate, multimodal feedback and enabling learners to self-correct during practice. The system's design significantly improves musical accuracy and learner engagement without requiring constant instructor supervision. The positive reception from both students and educators further validates the framework's practical utility and educational impact. SMART-TUNE addresses critical challenges in modern music education by offering scalable, personalized learning environments suitable for both remote and in-class settings. It bridges the gap between unsupervised practice and expert feedback, fostering continuous improvement and learner autonomy.

Future developments will focus on expanding instrument coverage, supporting polyphonic and genre-specific feedback, and integrating emotion recognition and biofeedback mechanisms. Additionally, collaborative cloud-based tools and gamified learning modules are planned to enhance engagement and support group-based instruction. These advancements aim to position SMART-TUNE as a comprehensive, data-driven platform for next-generation music education.

Author contributions

Material preparation, data collection and analysis were performed by He, The first draft of the manuscript was written by He, and He commented on previous versions of the manuscript.

Funding

No funding.

Data availability

All data generated or analysed during this study are included in this article.

Declarations

Ethics approval and consent to participate

This study did not require ethical approval as it did not involve human or animal subjects.

Consent to publish

Not applicable.

Competing interests

The authors declare no competing interests.

Received: 10 September 2025 / Accepted: 3 April 2026

Published online: 01 July 2026

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